

Advancing Today's Training Defeating Tomorrow's Threats

TSMO Threat Operations **EW In-Depth**





Threat Electronic Warfare Capabilities



MANEUVER AND NETWORK

Threat Battle Command Force (TBCF)

ELECTRONIC ATTACK

Enhanced Controlled Signal Transmitter (E-CST)
Cyber and Electronic Warfare Jammer Asset (CEWJA)
Cicada-C
Wideband Configurable Controlled Jammer (WCCJ)
TRC-274
Threat High Output Range Extender (THORE)
Mission Adaptable Multi-Band Asset for GNSS (MAMBA-G)

SIGINT / DF

Advanced Networked Electronic Support Threat Sensors (AdvNESTS)
Networked Electronic Support Threat Sensors (NESTS)
Electronic Warfare Open-air Collector (EWOC)

INTEGRATED AIR DEFENSE SYSTEMS

Giraffe Agile Multi-Beam (G-AMB)
Giraffe 1X (G1X)
Giraffe 75 (G75)
Anti-Aircraft Artillery Simulator (AAA2)
Radio Frequency Surface-to-Air Missile (RF SAM)
Radar Signal Emulator (RSE)

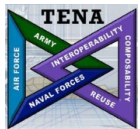
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Integrated Network Command and Control



Threat Battle Command Force (TBCF)



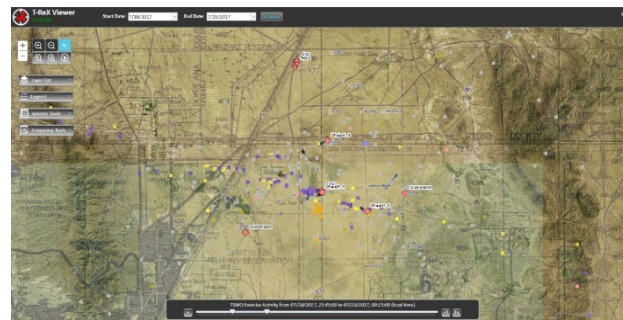
Built on Test and Training Enabling Architecture (TENA)

Allows easy translation and integration with other simulation languages

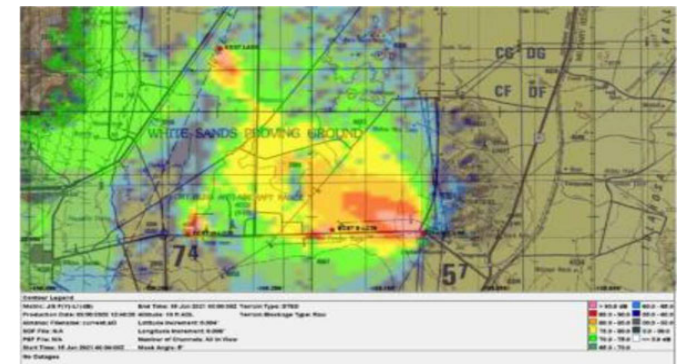
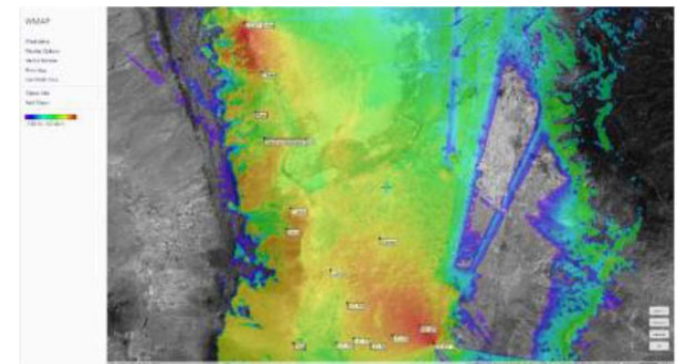
Adaptable Range Exercise System (ARES) Real-time operational Picture



TSMO Replay Exercise Viewer (T-ReX) Data recording and replay



RF Propagation Modelling and Analysis Spectrum-E (general purpose) and GIANT (GPS)



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Threat Battle Command Force (TBCF)



System Description

- Threat Command, Control and Communications Suite
- Integrated Threat Situational Awareness (SA) visualization
- Command and Control (C2)
- Common operating picture (COP) at multiple locations

Integrated with:

- Electronic Warfare (EW) assets
- Cyber Red Team
- Live, Virtual and Constructive (LVC) representations
- Traditional threat maneuver and fires assets

Components:

- ARES Common Operational Picture
- TReX Historical data collection and viewer
- TCP/IP network backbone over RF, copper, or fiber
- Data collection and analysis suite
- Test Enabling Network Architecture (TENA) interconnections
- Voice via Voice over IP (VoIP) or VHF / Instant messenger chat
- Shared Data Storage



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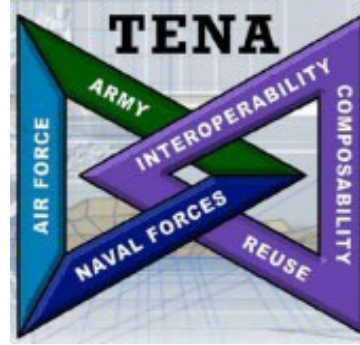


Visualization, Analysis, and Data Collection



Adaptable Range Exercise System (ARES)

- Produced by General Dynamics
- Real-time operational picture
- Visualization of:
 - Platform location and movement
 - NESTS DF tracks and LOB's
 - RADAR tracks
 - Jammer/EA activity
 - Sensor System Detections



Test and Training Enabling Network Architecture

- Enables Interoperability among systems, facilities, simulations and C4ISR systems
- Adapters translate platform data and tracks into common TENA Object Model Language
- Visualization/C2 tools can be easily duplicated anywhere on the TBCF network
- New or improved Systems can be easily integrated into existing TENA framework



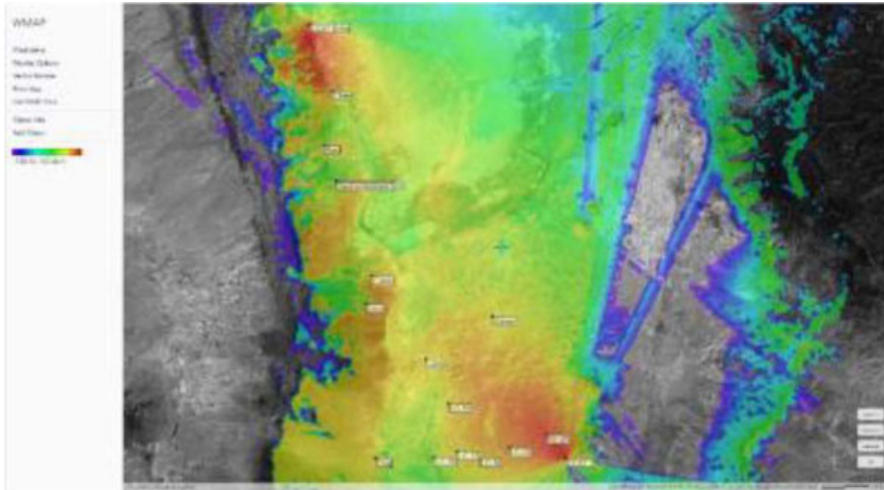
TSMO Replay Exercise Viewer (T-ReX)

- Based on ArcGIS, produced by ESRI
- Allows time-integrated viewing and filtering of:
 - NESTS DF tracks
 - Filter by Frequency
 - Color-code by identified unit
- Jammer fans

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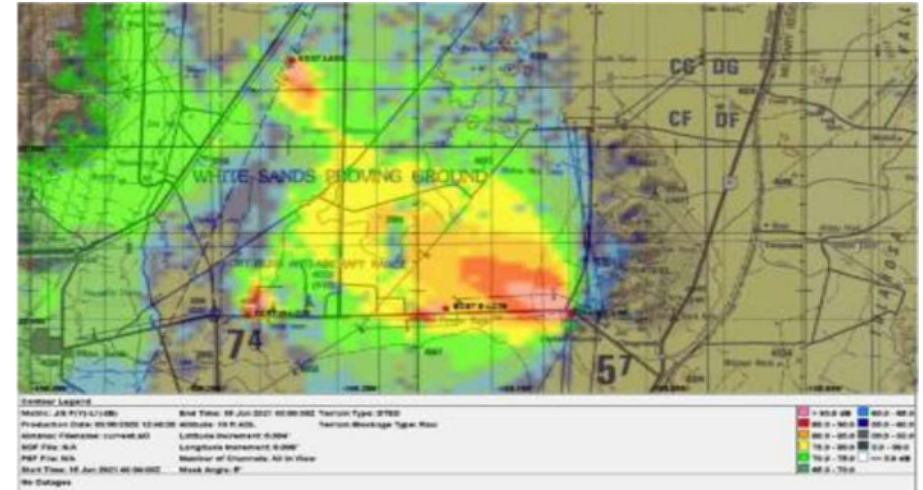


RF Propagation Modeling and Analysis



Spectrum-E / HTz Warfare

- Produced by ATDI
- Calculates:
 - Jammer power-on-target
 - Jammer-to-Signal Ratio
 - DF Miss Distance
 - Communications Signal Visibility Area / Power on Target
 - Communication Link Budget



GPS Interference and Navigation Tool (GIANT)

- US DoD-owned GPS analysis tool, produced by LinQuest
- Calculates Jammer-to-Signal and Repeater-to-Signal Ratios
- Used to obtain GPS emissions effects areas and permits

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Electronic Warfare Open-Air Collector (EWOC)



System Description

- RF detection and analysis laboratory
- Validates propagation modelling data for Electromagnetic Surveillance Vulnerability Assessment (ESVA) of BLUFOR units
- Provides ground truth on electronic warfare (EW) emissions
- Ensures compliance with FCC and Army guidelines for electronic attack (EA) emissions
- Confirms emitter direction (truck mounted only)



CONOPS

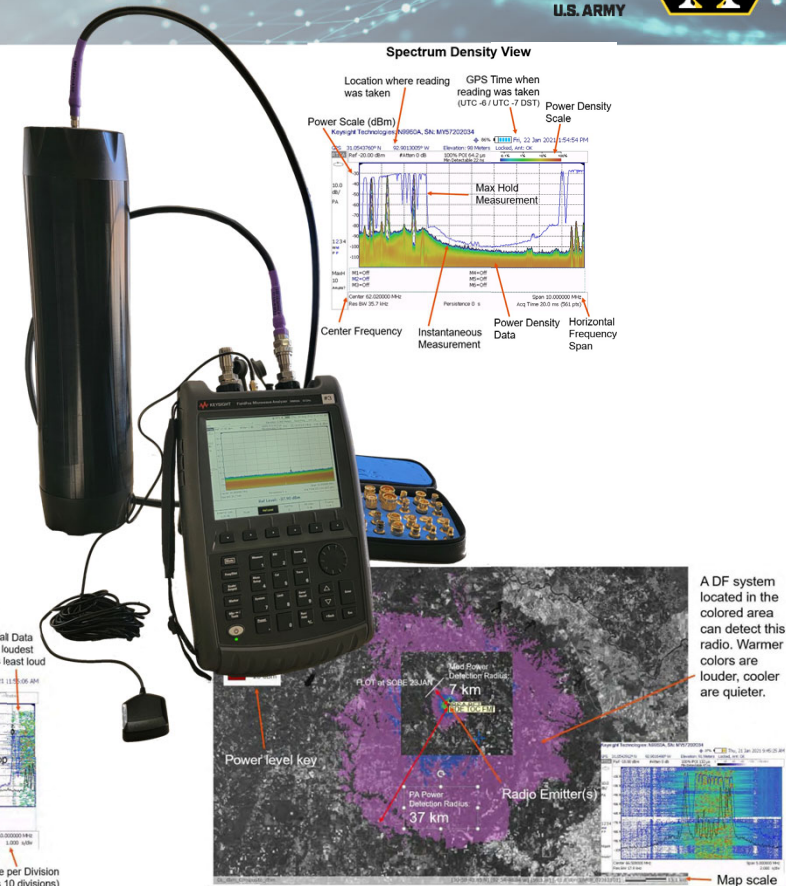
- Deployed near an EA system to verify emissions
- Deployed near system under test to verify EA power on target
- Used as a white-cell element to determine electromagnetic signature of target units, to assist in signature mitigation
- **NOT A THREAT REPRESENTATIVE SYSTEM**

EWOC Performance

- Real-time Signal Intelligence and Analysis
- Direction Finding
- Recording and playback of intermittent and pulsed signals
- Internal amplification of low-level signals

EWOC-P Performance

- Real-time Signal Intelligence and Analysis
- Internal amplification of low-level signals



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Advanced Networked Electronic Support Threat Sensors (AdvNESTS)



System Description

- Direction Finding (DF) using Angle and Time Difference of Arrival (AoA / TDoA)
- Frequency Correlation
- Signal Exploitation
- DF lines-of-bearing
- Emitter geo-location
- Net construction
- Signal analysis
- Provides an advanced communications detection infrastructure with Operational Coverage: 4 to 400 km², depending on terrain

CONOPS

- Provides electronic signal and DF data for the Threat Commander and Staff.
- Information displayed at EW TOC includes frequency, emitter location, detected power, track quality
- When correlated with known unit frequencies, yields Blue Force unit locations

Performance

- Dedicated digital DF receiver systems
- Robust digital radio network with redundant data options
- High capability against modern systems and waveforms including DSSS and FHSS signals



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Networked Electronic Support Threat Sensors (NESTS)



System Description

- Direction Finding (DF)
- Signal Exploitation
- Provides an advanced communications detection infrastructure. May be operated stand-alone or cooperatively networked.
- DF lines-of-bearing
- Emitter geo-location
- Net construction
- Signal analysis

Performance

NESTS-High

- Dedicated digital DF receiver systems
- Robust digital radio network with redundant data options
- High capability against modern systems and waveforms including DSSS and FHSS signals

NESTS-Medium

- Dedicated digital receiver only
- Digital data radio with redundant options
- Good capability against current tactical radios and cell phones

CONOPS

- Provides electronic signal and DF data for the Threat Commander and Staff.
- Information displayed at EW TOC includes frequency, emitter location, detected power, track quality
- When correlated with known unit frequencies, yields Blue Force unit locations



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Enhanced Controlled Signal Transmitter (ECST)



System Description

- Electronic attack (EA) of Communications and Unreliability – Current Threat
- Highly Mobile
- Capable of full remote operation through ITF network
- Remote system diagnostics
- Able to control Remote Jammer Units (RJUs) for Injection Jamming

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request a local or remote operator to conduct EA on a single simultaneous frequency.
- E-CST can find additional frequencies of interest in the area, resulting in electronic attack against those targets.

Performance

- Remote Controllable through ITF network
- Setup time: 30 min - 2 hrs.



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Cyber and Electronic Warfare Jammer Asset (CEWJA)



System Description

- Software Defined Jammer
- Reprogrammable, multi-functional EA support
- Smart electronic attack
 - ESM + ECM focuses emitted power on the targets of interest to optimize the jamming effects
- Fast triggering and wide band receiving
- Effectively jams all kinds of modern communications, including frequency hopping combat net radios

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request CEWJA conduct Electronic Attack (EA) on multiple frequencies in different bands
- CEWJA can find additional frequencies of interest in the area resulting in EA against those targets
- CEWJA can automatically jam frequencies upon detection

Performance

- Smart ESM and ECM capabilities
- Multirange: HF, V/UHF, SHF, and C-Band
- Library of jamming waveforms
- Custom waveform creation software
- Follower-jammer
- DRFM repeater



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Cicada-C



System Description

- Jams up to 16 active radio nets simultaneously
- Example targets: Frequency hoppers, cell phones, satellite navigation
- Reactive jam modes scan the environment and activate only when target signals are present

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request CICADA conduct Electronic Attack (EA) on multiple frequencies in different bands
- CICADA can find additional frequencies of interest in the area resulting in EA against those targets
- CICADA can automatically jam frequencies upon detection

Performance

- Automatic, computer controlled jamming sequences
- High power, liquid cooled power amplifiers
- Extremely fast multiplex operation for effective jamming of multiple targets simultaneously
- Special jamming modulations ensuring effectiveness
- Jamming of predefined frequencies in order of priority
- Barrage jamming against several frequency hoppers
- Protected frequencies to be spared from jamming
- Low false alarm rate and high probability of intercept
- Setup time: 1-2 hours



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Wideband Configurable Controlled Jammer (WCCJ)



System Description

- EA of Communications and Unreliability – Current Threat
- Example Targets: *FM Voice, SRW / WNW, Satellite Communications*
- Minimal out-of-band emissions, even near restricted frequencies
- Replicates many different Threat jamming systems

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request the WCCJ to conduct electronic attacks (EA) on 2 different frequencies simultaneously.
- WCCJ can find additional frequencies of interest in the area resulting in EA against those targets.

Performance

- Frequency follower plus other Direct Sequence Spread Spectrum (DSSS) threat jamming capabilities
- Digital Radio Frequency Memory (DRFM) capability
- Direction Finding (DF) capability
- Wideband frequency output
- 2 independent output channels allow simultaneous jamming of 2 separate frequencies
- Dynamically Programmable Output Waveforms
- Flexible output frequency tailoring
- Remote monitor capability using the ITF network
- Setup time: 1-2 hours



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TRC-274 Software Defined Radio Jammer



System Description

- Software Defined Jammer
- Reprogrammable, multi-functional EA support
- Smart electronic attack
 - ESM + ECM focuses emitted power on the targets of interest to optimize the jamming effects
- Fast triggering and wide band receiving
- Effectively jams all kinds of modern communications, including frequency hopping combat net radios

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request TRC-274 conduct Electronic Attack (EA) on multiple frequencies in different bands
- TRC-274 can find additional frequencies of interest in the area resulting in EA against those targets
- TRC-274 can automatically jam frequencies upon detection

Performance

- Smart: ESM and ECM capabilities
- Fast trigger time
- Multirange: HF, V/UHF, and SHF
- High performance
- Library of jamming waveforms
- Spot and Responsive jamming
- Deception
- Smart chirp
- Follower-jammer
- DRFM repeater
- Setup time: 1-2 hours



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Threat High Output Range Extender (THORE)



System Description

- Unmanned Network Controlled PNT Jammer System
- Electronic attack (EA) of PNT signals using current threat techniques
- Capable of full remote operation through TBCF network
- Remote system diagnostics

CONOPS

- Sophisticated, programmable radio frequency (RF) threat emulator
- Provides high fidelity simulation of known Position, Navigation and Timing (PNT) denial threats
- Designed to assess operational capabilities in Global Navigation Satellite Systems (GNSS) degraded/denied environments

Performance

- Remote Controllable through TBCF Network
- Setup time: 30 min – 1 hr.



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Mission Adaptable Multi-Band Asset for GNSS (MAMBA-G)



System Description

- Unmanned Network Controlled PNT Jammer System
- Electronic attack (EA) of PNT signals using current, emerging, and currently-emerging threat techniques
- Capable of full remote operation through TBCF network
- Remote system diagnostics

CONOPS

- Sophisticated, programmable radio frequency (RF) threat emulator
- Provides high fidelity simulation of known Position, Navigation and Timing (PNT) denial threats
- Designed to assess operational capabilities in Global Navigation Satellite Systems (GNSS) degraded/denied environments

Performance

- Remote Controllable through TBCF Network
- Setup time: 30 min – 1 hr.



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Giraffe Agile Multi-Beam (GAMB)



System Description

- Multi-beam 3-Dimensional air coverage
- Frequency agile pulse Doppler air search radar
- Mobile or static short to medium range air defense application
- Detects hovering helicopters
- Detects in-coming artillery and mortar fire
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS

- Air surveillance and tracking
- Fixed-wing and Rotary-wing tracking
- Rocket, Artillery, and Mortar Tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- All tracks displayed real-time on COP via ITF network

Performance

- Multi-beam 3-Dimensional air coverage with instrumented ranges
- Operational Frequency range
- Data rate 1-scan-per-second
- Pulse density suppresses high cluttering in adverse weather.
- Ultra-low antenna side-lobes combined with pulse-to-pulse and burst-to-burst frequency agility provides some resistance to jamming
- Automatic Point of Origin / Point of Impact calculations
- Deployable in 10 minutes, prepared to travel in 6 minutes.



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Giraffe 1X (G1X)



System Description

- Newest member of the Giraffe family of radar
- Automated Threat Evaluation and Weapons Assignment
- Target data and fire control orders for allocated weapons
- Distribution of target data over tactical data links
- Detects hovering helicopters
- Detects in-coming artillery and mortar fire
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS

- Air surveillance and tracking
- Fixed-wing and Rotary-wing tracking
- Rocket, Artillery, and Mortar Tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- All tracks displayed real-time on COP via ITF network

Performance

- Multi-beam 3-Dimensional air coverage
- X Band, Multi Channel
- Simultaneous tracking of surface and air tracks
- Data rate 1-scan-per-second
- Pulse density suppresses high cluttering in adverse weather.
- Ultra-low antenna side-lobes and frequency hopping with auto selection of least jammed frequency



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Giraffe 75



System Description

- Frequency agile pulse Doppler air search radar
- Mobile or static short to medium range air defense application
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS

- Air surveillance and tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- Can provide an air picture to each firing battery using FM radio communication

Performance

- 13-meter antenna mast
- Includes an integrated Command, Control and Communication (C3) system that enables GIRAFFE to act as the command-and-control center in an air defense system



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Anti-Aircraft Artillery Simulator (AAA2)



System Description

- Ground-based air defense threat emitters
- Create a realistic electronic warfare environment for training fixed- and rotary-wing aviators
- Daylight and FLIR cameras for target acquisition, RADAR pulse emitter to stimulate RWS

Emulated 23-mm SP Anti-aircraft Artillery

Day and Night Vision

On-Board Radar

- Detection
- Tracking

23 mm AA guns

Emulated 30-mm SP Anti-aircraft Artillery

Day and Night Vision

Dual Tracking / Detection Radars

2 x 30 mm twin barrel cannons

- *Surface-to-air missiles*
- Must stop to fire missiles



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Radio Frequency Surface-to-air Missile Simulator (RF-SAM)



System Description

- Ground-based air defense threat emitters
- Create a realistic electronic warfare environment for training fixed- and rotary-wing aviators
- Daylight and FLIR cameras for target acquisition, RADAR pulse emitter to stimulate RWS

Armament

Surface-to-air missile

Radars

Target Acquisition Radar

Mono-pulse Target Tracking Radar

Missile Tracking



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Thank you